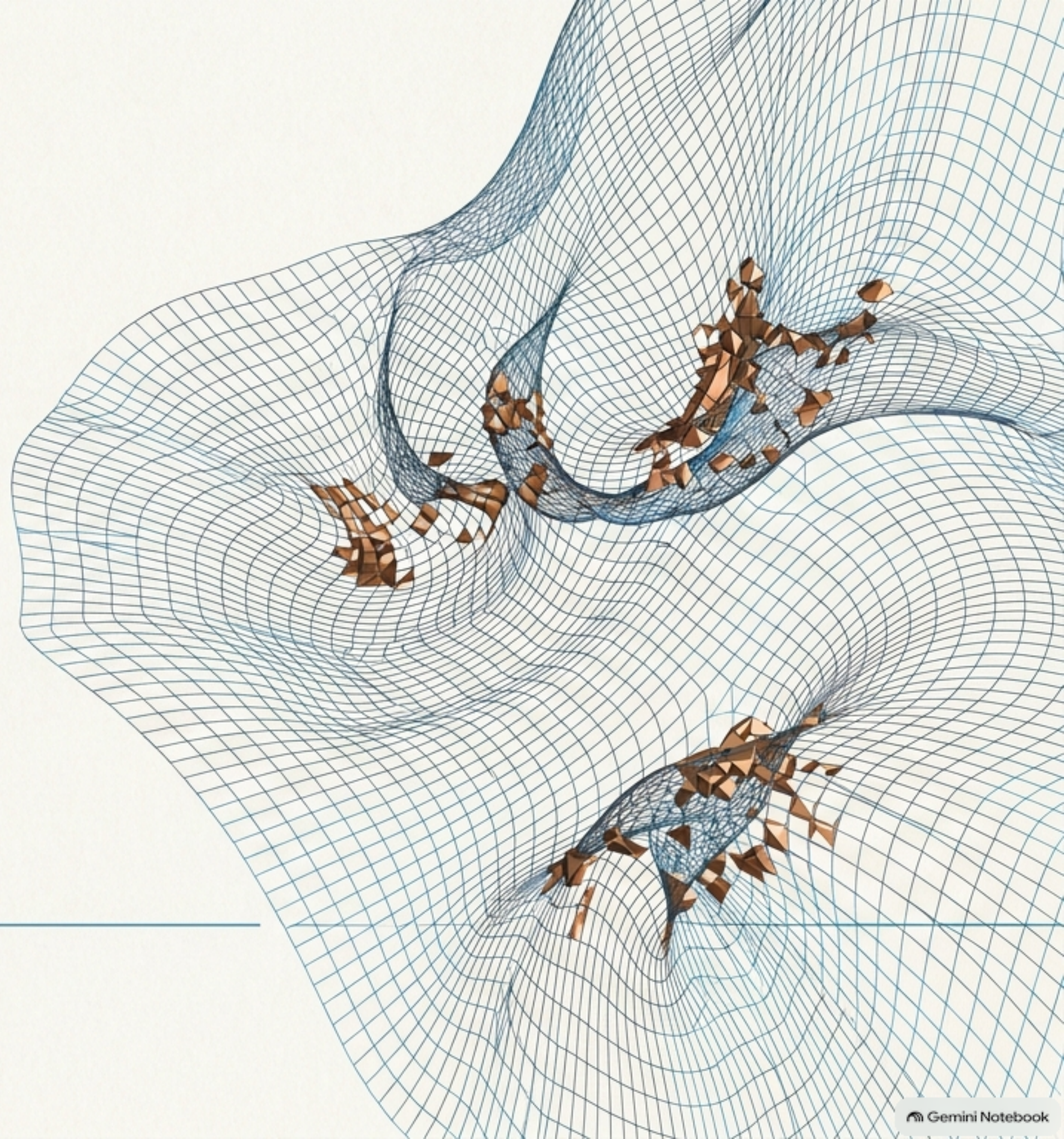


Gravity as Entropic Descent

Mapping the Relativistic
Scalar-Vector Plenum
(RSVP) Framework

A state-of-the-research briefing on volume-filling entropic gravity, coherent polarization, and simulation provenance.



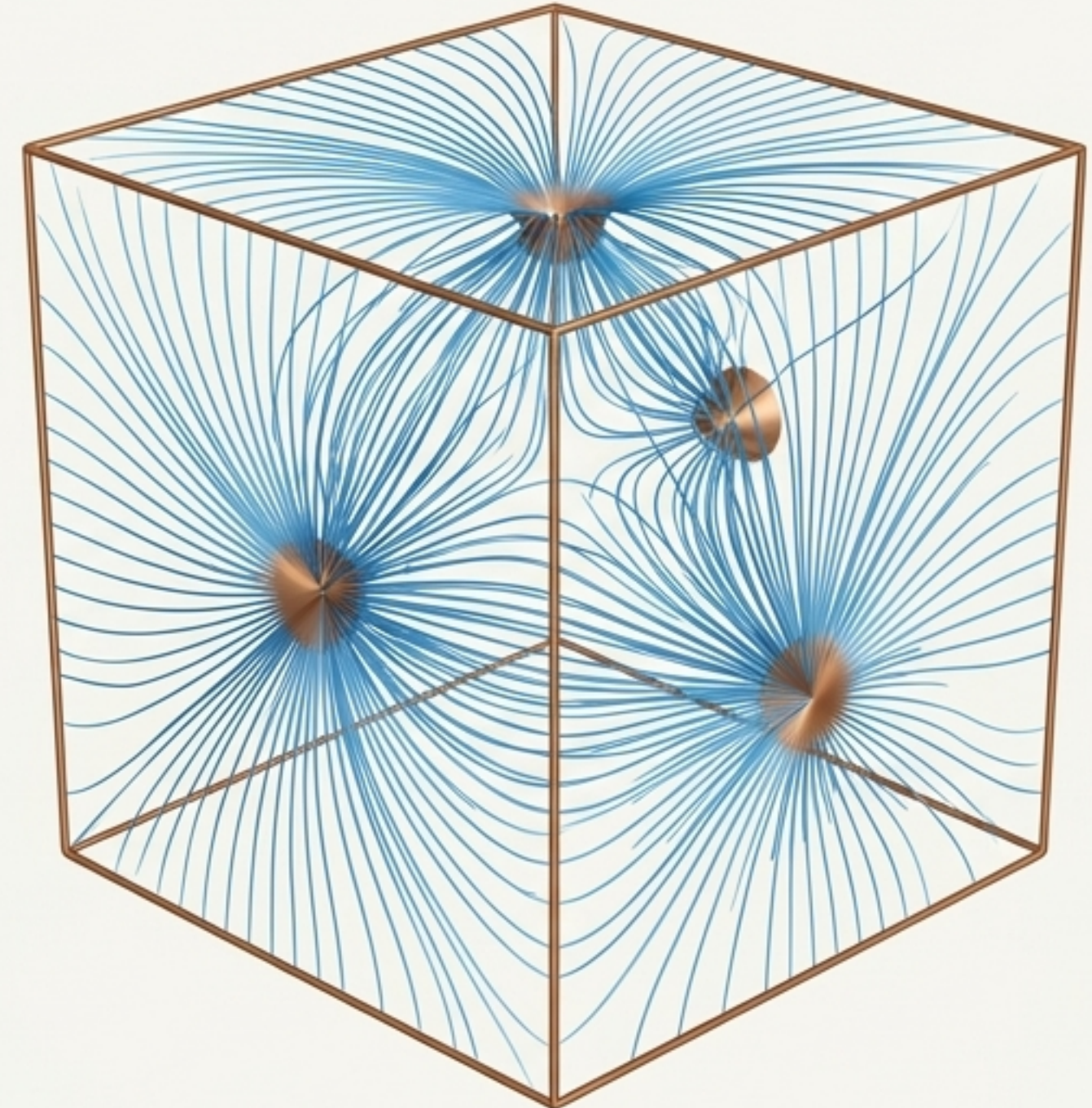
The Entropic Gravity Landscape

Theory / Proponent	Entropy Location	Core Mechanism	Key Result
Jacobson (1995)	Local Causal Horizon (Boundary)	Clausius relation ($\delta Q = T \, dS$)	Derivation of Einstein equations from thermodynamic state.
Verlinde (2010, 2016)	Holographic Screen & de Sitter Horizon	Entropic force & elastic media response	Derivation of Newton's laws; Apparent dark matter scaling.
Apparent-Horizon Cosmology	Cosmological Apparent Horizon	First law of thermodynamics at boundary	Derivation of Friedmann equations for cosmological evolution.
RSVP Framework	Volume-Filling Bulk (Plenum)	Coupled relaxation & entropic descent	Modified Poisson/field equations from internal field dynamics.

Relocating the Degrees of Freedom



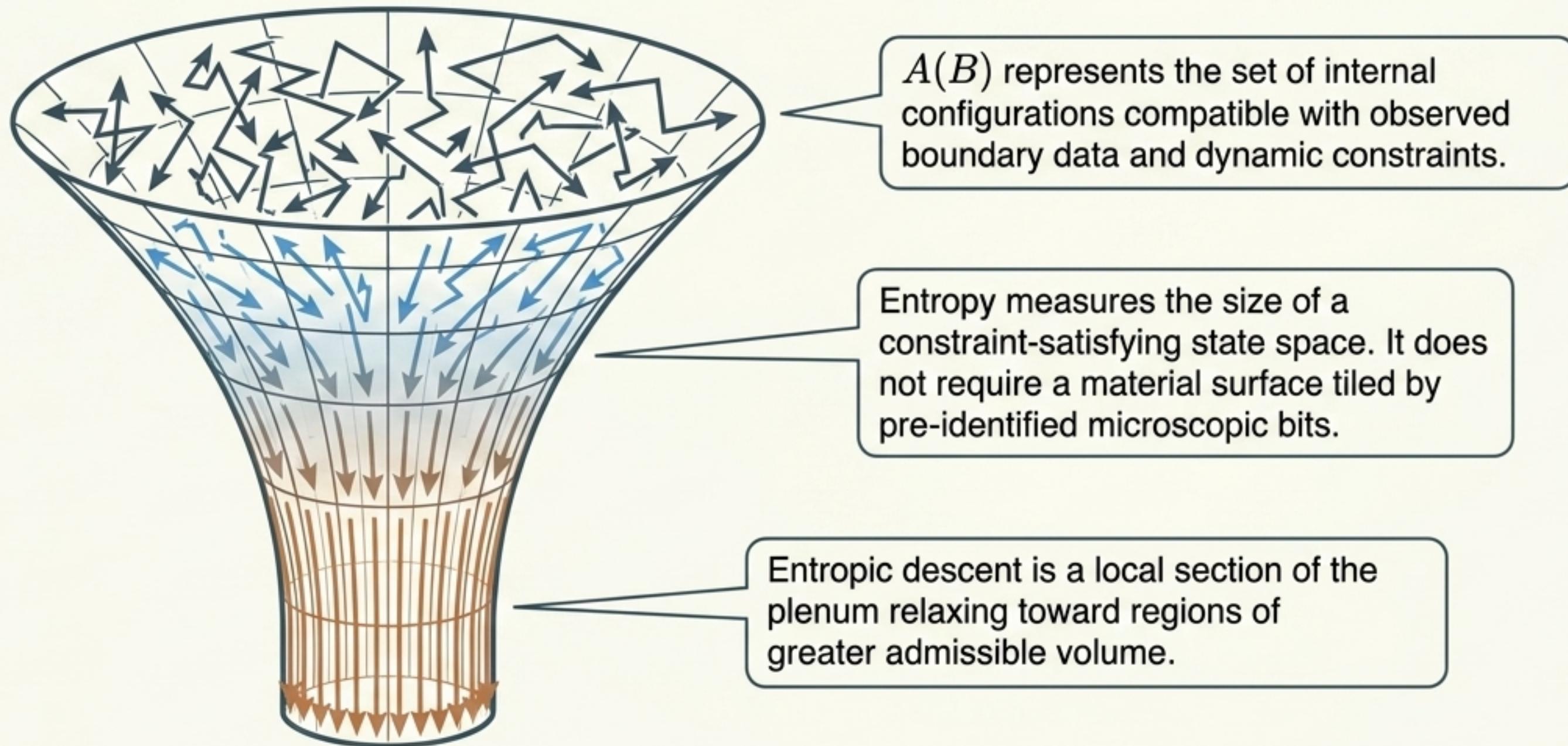
Horizon Thermodynamics constrains dynamics at a boundary. Gravity emerges as a force applied to matter from a distinguished screen.



The RSVP Plenum assigns a scalar (Φ), vector ($\mathbf{v}$), and entropy (S) field directly to every point in a volume. Gravity emerges as the coherent macroscopic consequence of local relaxation.

Admissibility as Gravitational Entropy

$$S(B) \sim \log \text{Vol}(A(B))$$



The Variational Formulation of the Plenum

[EPISTEMIC TAG: FORMAL THEOREM]

Core Fields: $\Psi = (\Phi, \mathbf{v}, S)$

Constraint Structure:
Divergence of the flow
field relates to scalar
and entropy sectors.

$$\nabla \cdot \mathbf{v} \approx \alpha_\Phi \Phi + \alpha_S S$$

Modified General Relativity:
 $\Theta_{\mu\nu}$ collects entropy-gradient
contributions like $\nabla_\mu S \nabla_\nu \Phi$.

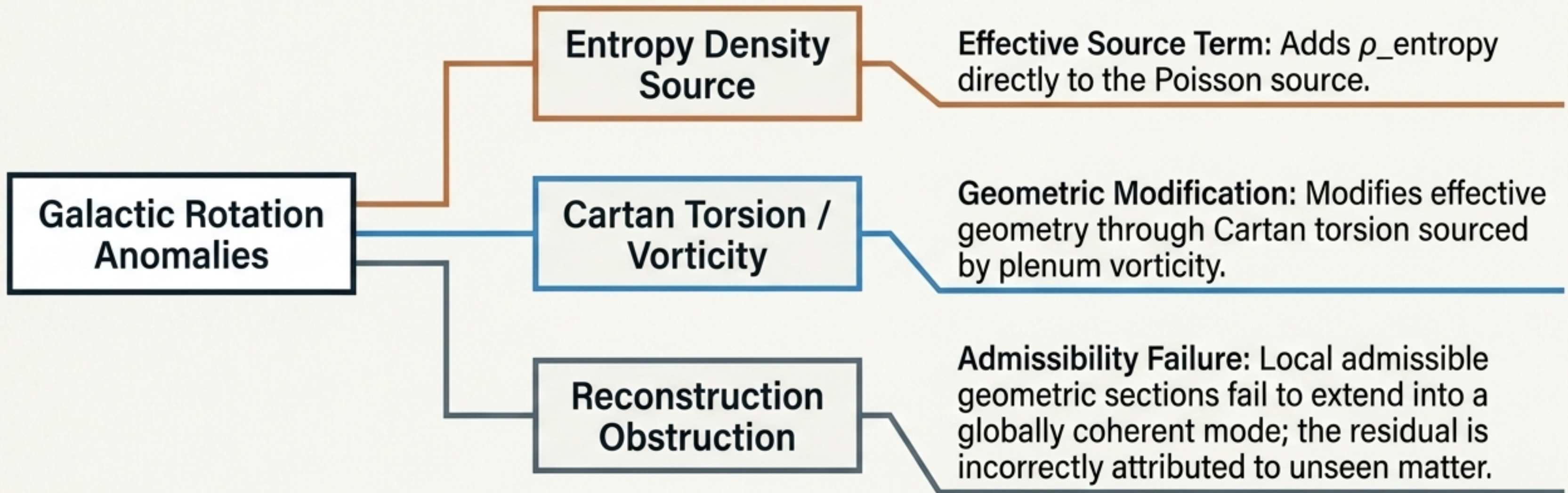
$$G_{\mu\nu} + \Theta_{\mu\nu} = 8\pi G T_{\mu\nu}$$

Weak-Field Poisson Limit.

$$\nabla^2 \Phi_{\text{grav}} = 4\pi G (\rho_{\text{matter}} + \rho_{\text{entropy}})$$

Key Takeaway: Deriving the equations from an action principle is an established theorem. Interpreting ρ_{entropy} as a physical density is a phenomenological claim.

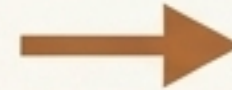
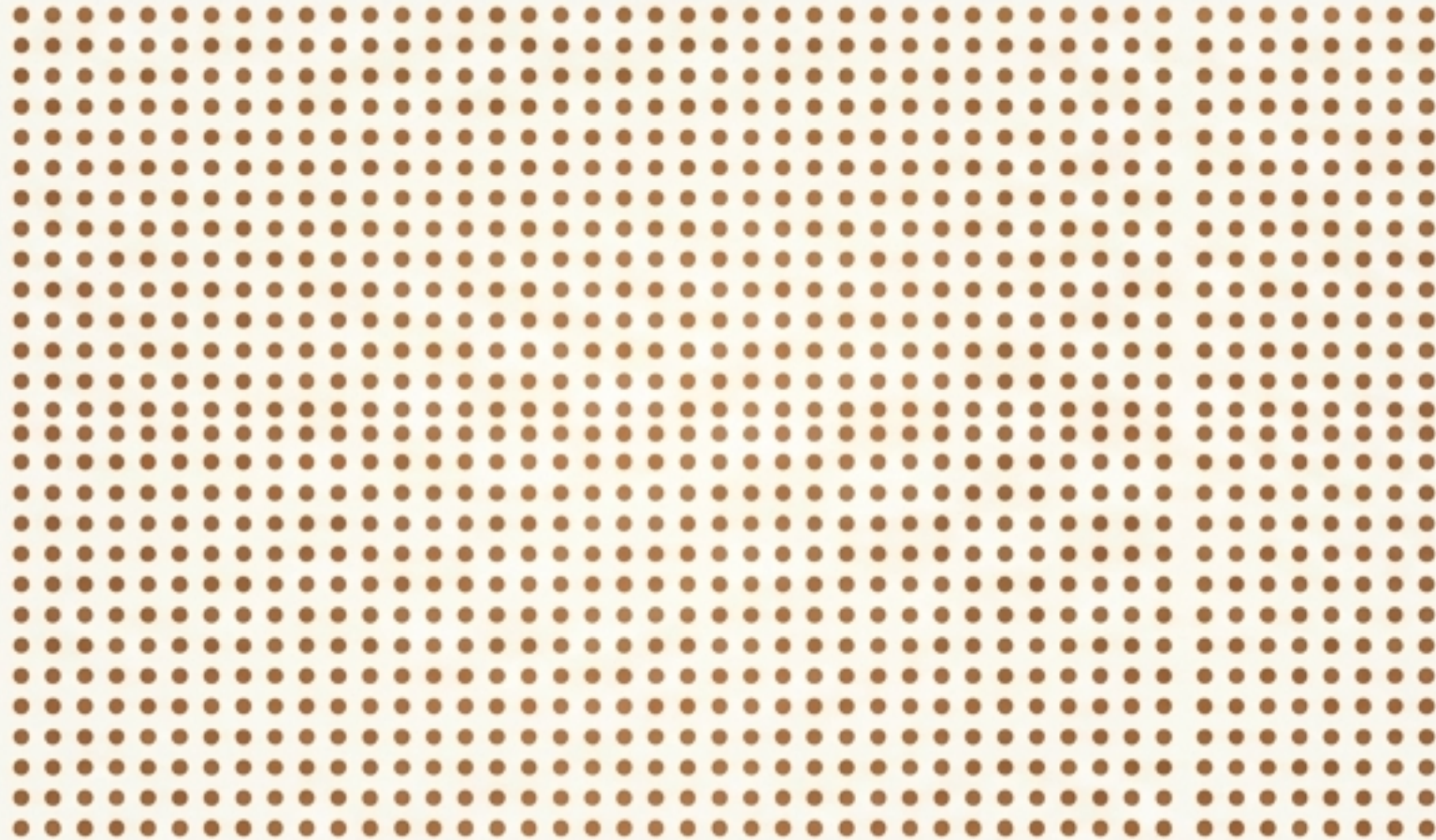
Dark Matter Phenomenology: Plural, Not Unified



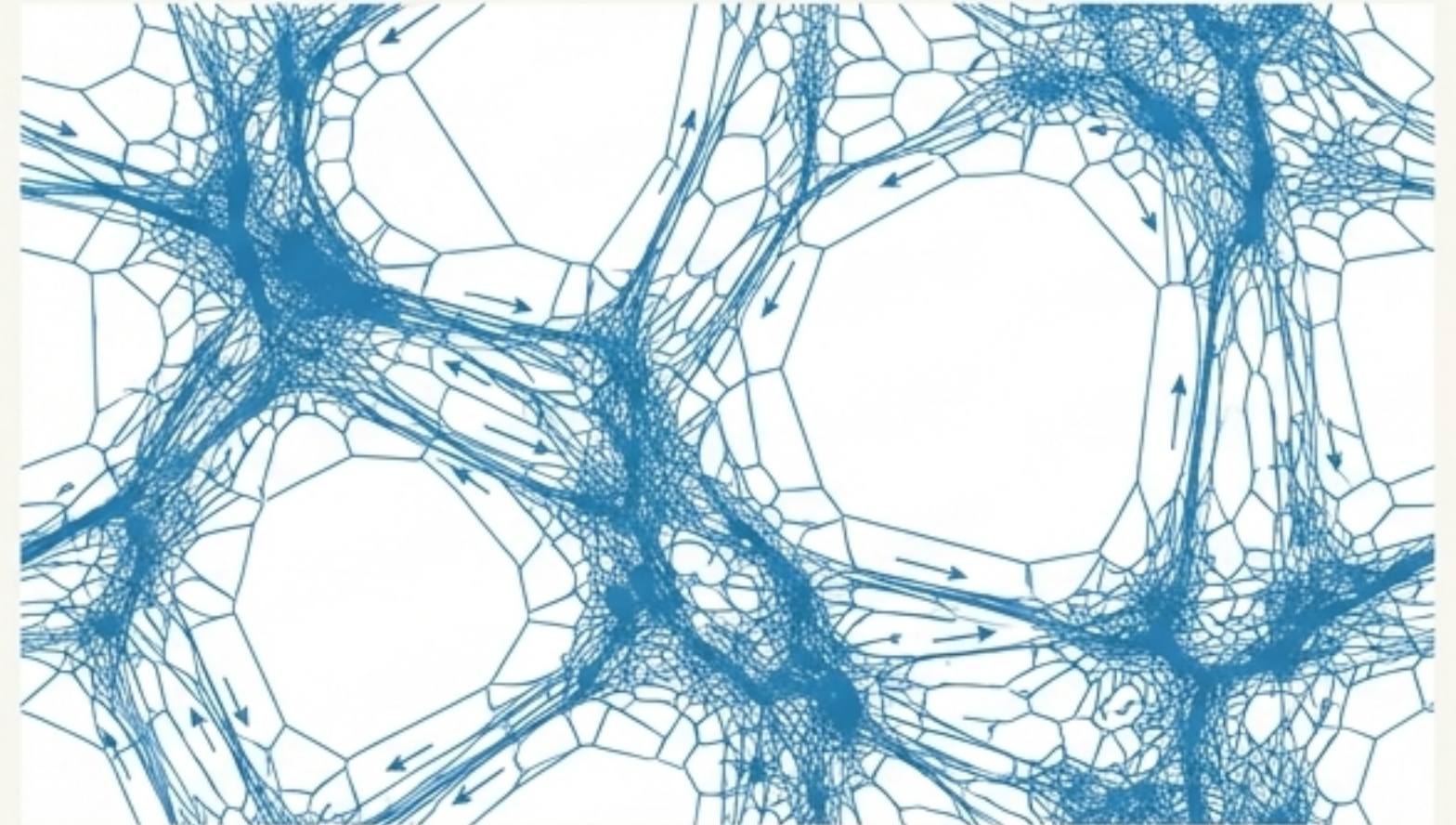
Note: These are mathematically distinct objects. They are not yet proven to yield the same lensing map, rotation curve, or cluster dynamics.

Cosmic Evolution: From Brick to Sponge

Brick



Sponge



Uniform metric expansion is replaced by entropy redistribution, scalar permeability, and vector flow as the primary explanatory vocabulary.

Cooling hydrogen forms Voronoi-like structural clustering.

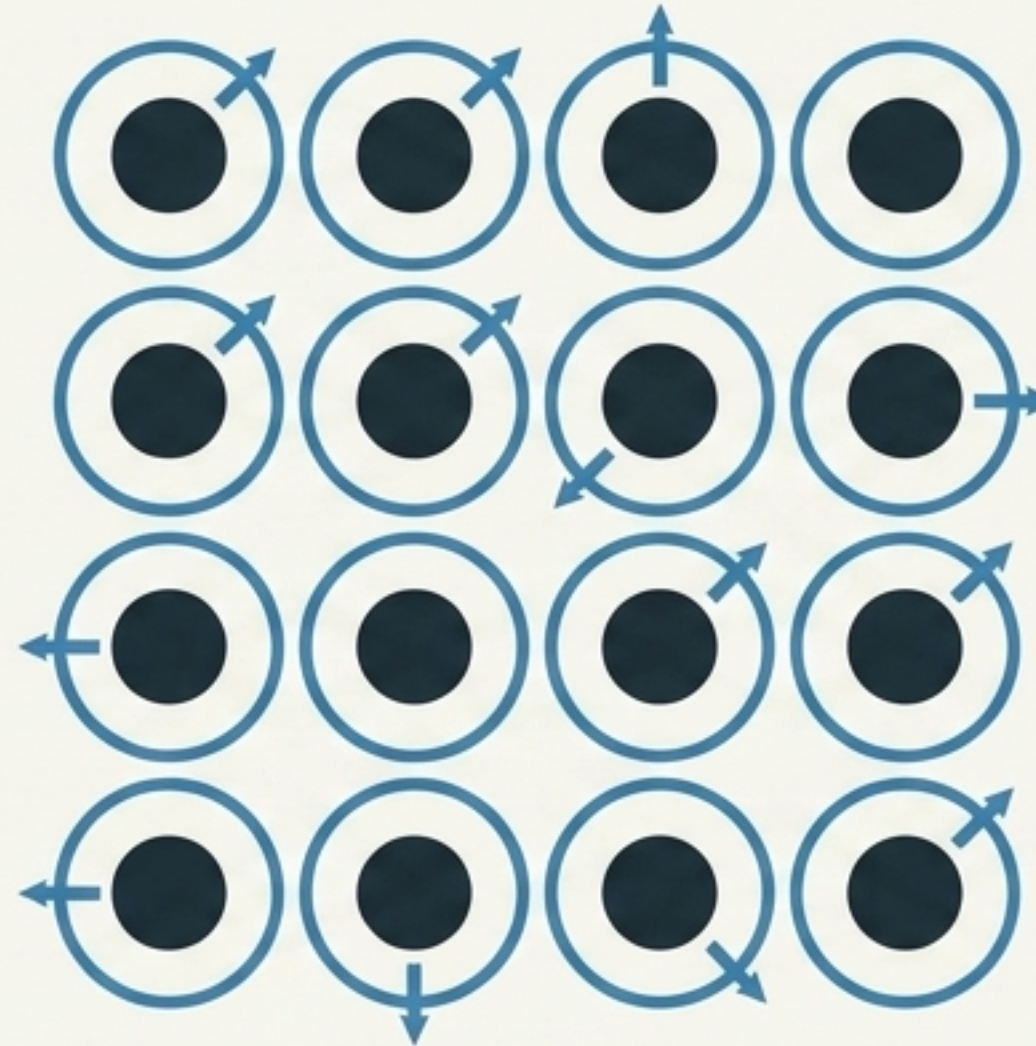
Lamphrodyne Circulation: Asymmetric inflow along the network supplies angular momentum to collapsing regions, transporting energy without requiring separately postulated initial rotation.

Coherent Polarization Without Collapse

A population of locally coupled elements acquires macroscopic coherence without erasing individual history.

The Micro:

Each element i carries a persistent, private internal state $x_i(t)$ and an exposed coordinate $\theta_i(t)$ for local coupling.

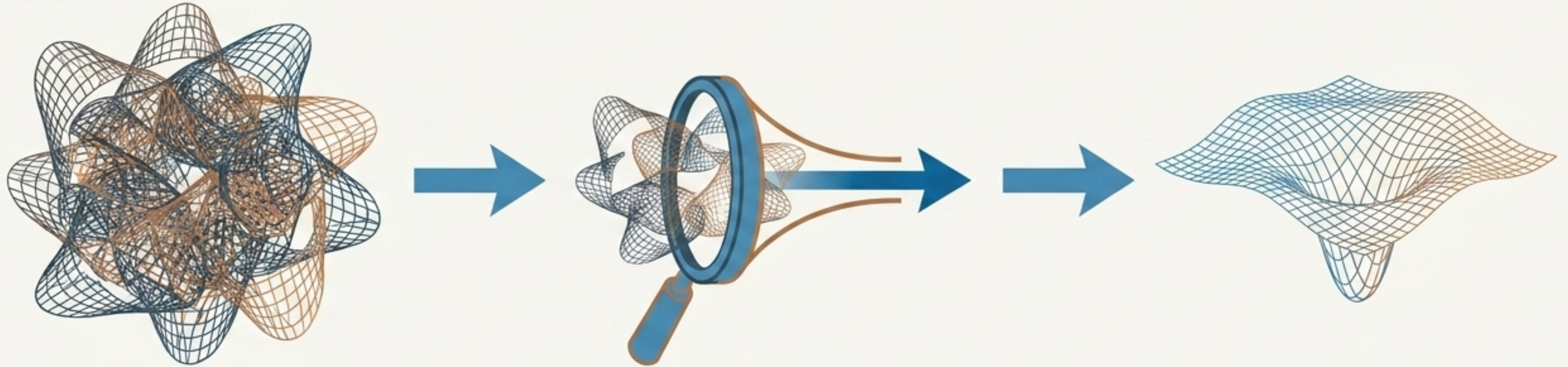


The Macro: Order parameter

$R e^{i\Psi} = (1/N) \sum e^{i\theta_i}$
emerges globally. No element broadcasts or receives a description of the full global state.

This mathematically licenses treating plenum polarization as spontaneous symmetry breaking, without claiming gravity is literal electromagnetism.

The Magnetization Bridge



Local States

$(\Phi_i, \mathbf{v}_i, S_i)$. The raw scalar, vector, and entropy data at each local point in the plenum.

Coarse-Graining

$\mathbf{m}(\mathbf{x}, t)$. A collective polarization field obtained by coarse-graining local configurations (e.g., $\mathbf{m}_v = \langle \mathbf{v} \rangle_\ell$).

Gravitational Potential

Φ_{grav} . The effective coupling, governed by $\nabla^2 \Phi_{\text{grav}} = 4\pi G \rho_{\text{matter}} + \lambda I[\mathbf{m}]$.

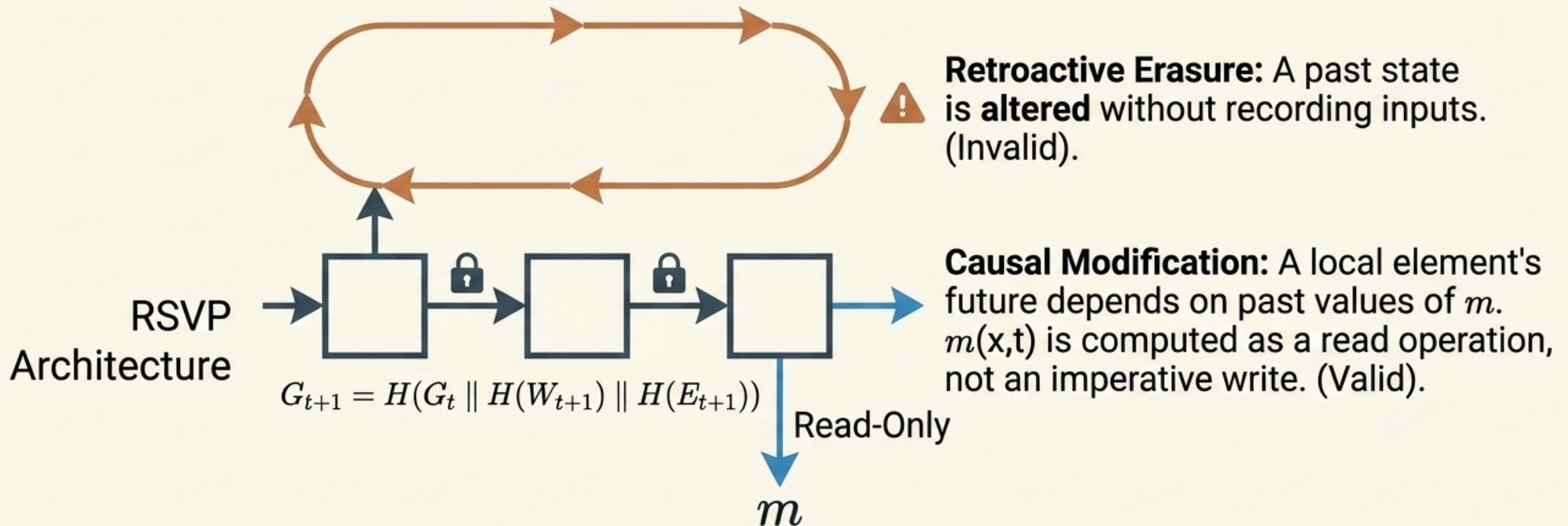
$\mathbf{m}(\mathbf{x}, t)$ acts as a deferred symbolic quantity bridging local entropic alignment to the observed gravitational metric.

From Analogy to Effective Theory

Established Components	Pending/Missing Components
✓ Microscopic Variables: Local plenum elements ($\Phi_i, \mathbf{v}_i, S_i$) are available.	⚠ Microscopic Spin: The reduced degree of freedom acting as a 'spin' is not yet uniquely identified.
✓ Observable Order Parameter: $m(\mathbf{x}, t)$ is proposed; definition and map to observables in progress.	⚠ Interaction Functional: Effective Ising or vector-spin Hamiltonian has not been derived from the RSVP Lagrangian.
	⚠ Symmetry Breaking: Relevant symmetry group and broken phase remain to be demonstrated.
	⚠ Control Parameter: A critical point for the transition has not been tied to a measurable quantity.

Simulation Provenance & Replayable History

The Threat: If $m(x,t)$ feeds back into local dynamics, does it overwrite the history that produced it?

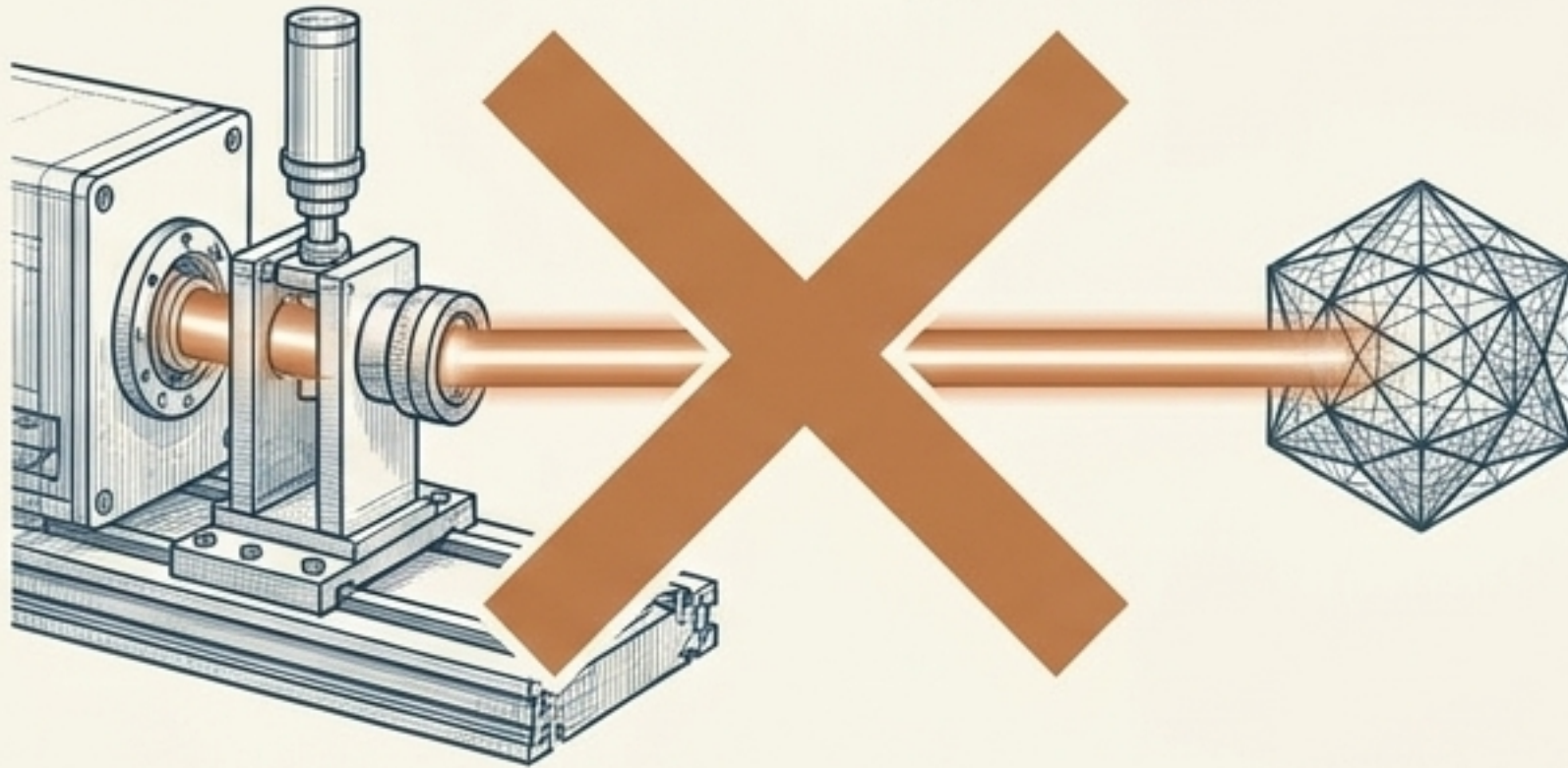


Ensures deterministic replayability from initial conditions without retroactive erasure.

What Emergence Does Not License

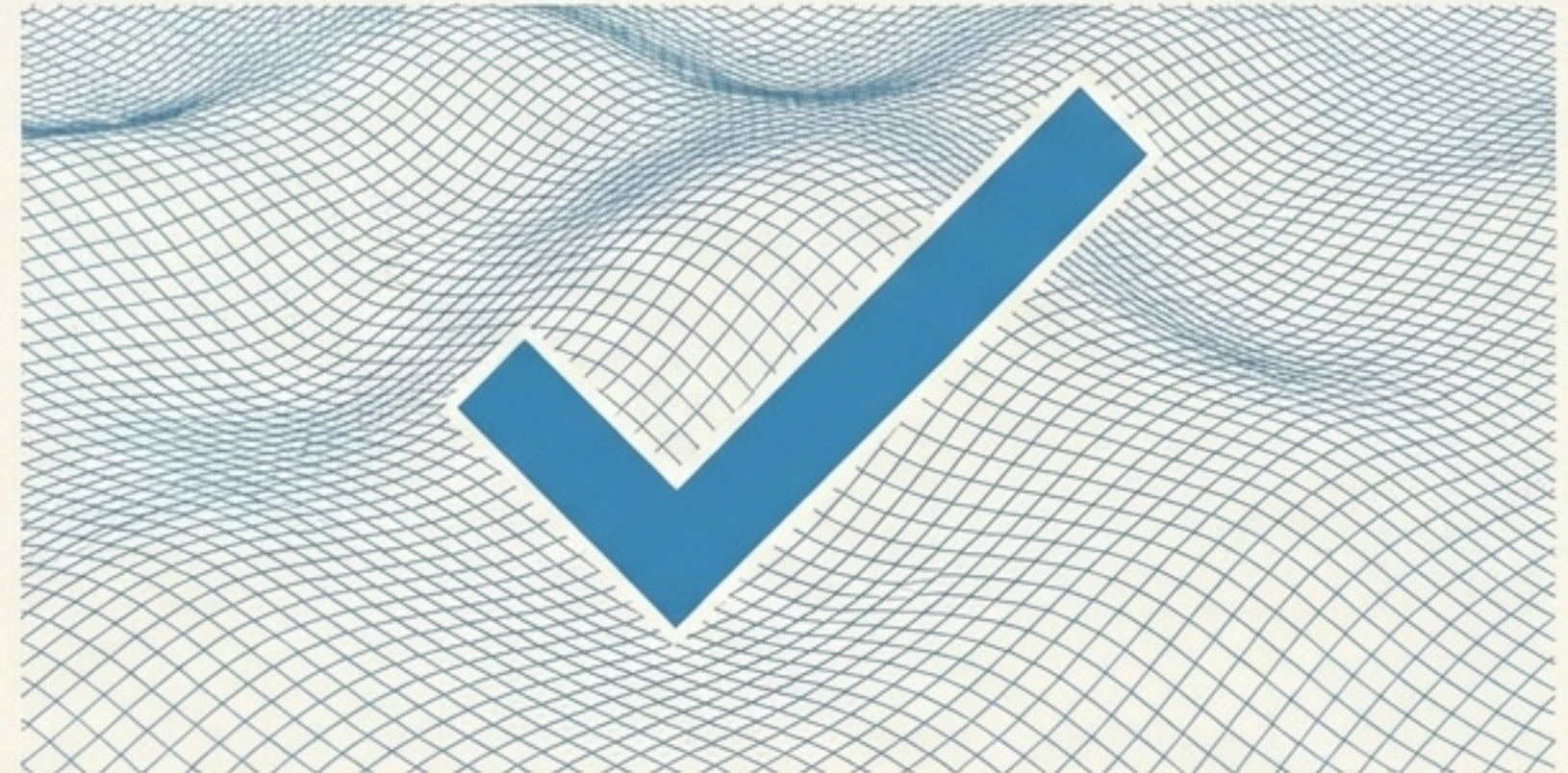
A thermodynamic derivation explains why macroscopic equations follow from collective constraints. It does not imply local controllability.

No Anti-Gravity



Local field content does **not** establish local **engineering**. Controlling the underlying state space requires a specified coupling to matter, an energy budget, and causal dynamics—none of which are granted automatically.

The Metaphor Limit



Universality preserves long-distance scaling properties; it does **not** preserve material **identity**. Gravity may share a statistical class with ferromagnetism, but it is not literal electromagnetism.

The RSVP Map: A Synthesized Territory

Theoretical Core

Relocating entropic descent from the boundary to the bulk, deriving general relativity limits through local constraint satisfaction.

Phenomenological Frontiers

Utilizing entropy density, Cartan torsion, and structural clustering to provide testable, plural alternatives to dark matter and dark energy.

Computational Discipline

Anchoring coherent polarization ($\mathbf{m}(\mathbf{x}, \mathbf{t})$) in deterministic, hash-chained simulations to guarantee physical replayability without retroactive erasure.

Gravitation modeled as the coherent macroscopic orientation of locally situated relaxation dynamics.